A Grid of Numbers · IG · Practice

U09b · 24 written + 6 pen-first = 30 marks · Name: _____

Fresh authored questions. Show working for calculations. Unless stated otherwise, calculate packed, uncompressed pixel data only: no header, palette, padding or compression. 1 byte = 8 bits; 1 KiB = 1024 bytes; 1 MiB = 1024 KiB.

1. Ours: what is represented?

(a) State what is meant by a pixel in a bitmap image. **[1]**

(b) State what is meant by image resolution. **[1]**

(c) State what is meant by colour depth. **[1]**

(d) State the maximum number of colours representable using 5 bits per pixel. **[1]**

2. Ours: two units for depth

(a) A game portrait is 384 × 256 pixels at 16 bits per pixel. Calculate its pixel-data size in KiB. Show working. **[2]**

(b) A museum label image is 350 × 240 pixels at **3 bytes per pixel**. Calculate its pixel-data size in bytes. Show working. **[2]**

A Grid of Numbers · IG · Practice

U09b · 24 written + 6 pen-first = 30 marks · Name: _____

3. Ours: choose a smaller image

A newly captured photograph is 960 × 640 pixels at 24 bits per pixel. Each option below changes only the stated setting.

(a) Reduce the width to 480 pixels, keeping height and colour depth unchanged. State the factor by which pixel-data size changes and explain why. **[2]**

(b) Instead, reduce colour depth to 8 bits per pixel, keeping both dimensions unchanged. State the size factor and one possible visible cost when encoding the photograph. **[2]**

(c) An editor changes an existing 8-bit image to a 24-bit representation without new source data. Explain why this does not necessarily restore colour differences lost earlier. **[2]**

4. Ours: store a colour

Explain how binary values represent the colours of pixels in a bitmap image. **[2]**

A Grid of Numbers · IG · Practice

U09b · 24 written + 6 pen-first = 30 marks · Name: _____

5. Ours: the bytes do not match

(a) A camera image contains **2 097 152 pixels**, each using 24 bits. Calculate the uncompressed pixel-data size in MiB. Show working. **[2]**

(b) Its saved file is smaller than your answer. Suggest one reason and explain why this does not make the pixel-data calculation wrong. **[2]**

6. Ours: a sign for a small display

(a) A display asset must fit a 64 KiB pixel-data budget. A designer proposes 256 × 256 pixels at 16 bits per pixel. Calculate its size in KiB and decide whether it fits. **[2]**

(b) Keeping colour depth unchanged, propose new dimensions that use exactly half as many pixels. Explain one possible cost of this reduction. **[2]**

A Grid of Numbers · IG · Practice

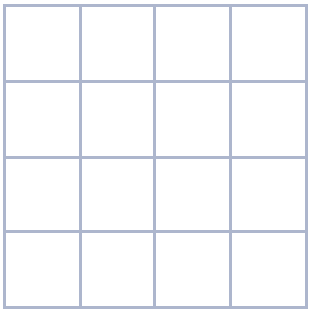
U09b · 24 written + 6 pen-first = 30 marks · Name: _____

7. Ours: reconstruct a tiny picture · pen first [6]

💻 **Design, then trace.** No code is required. A 4 × 4 one-bit bitmap stores rows from top to bottom, each row left to right. 1 means black; 0 means white. Its packed pixel bits are **1000 0100 0010 0001**. These six marks are human marked.

(a) Write numbered steps to turn the supplied bits into a picture. Include when to start a new row. **[2]**

(b) Trace your steps: shade the grid to reconstruct the picture. **[2]**



(c) A separate RGB pixel has three 8-bit channels: R = 12, G = 128, B = 225. Write its six-digit hexadecimal colour as #RRGGBB. Preserve leading zeroes. **[2]**

Above and beyond · outside the 30 marks. Two 2 × 2 black/white patches have rows **01 / 10** and **00 / 11**. A resizer replaces each patch with its mean value (black = 1, white = 0). Predict the two outputs. Can that single output uniquely identify which original patch was used? Explain.

Additional space - label the question number.